EMERBRACE

A smart emergency bracelet for instant emergency alerts.

A portable and affordable Smart emergency alert bracelet

“Save lives through instant connectivity.”

PROBLEM

- Chronic illnesses: **71%** of global deaths; **85%** in low/middle-income countries.
- Diabetes: **4.1 million deaths** per year globally; many preventable with timely care.
- Heart attack: **17.9 million deaths** per year globally; majority in low-resource areas.
- Stroke: **33%** die; **33%** permanently disabled
- Pregnant women: **70% of maternal deaths** due to delayed help; **94%** occur in low-resource settings.

OBJECTIVE

- Provide **affordable**, life-saving tech **in low-resource areas** like Ethiopia.
- Empower high-risk patients with real-time alerts for seizures, pregnancy emergencies, and sudden health events.
- Reduce preventable deaths from chronic illnesses, maternal pregnancy deaths, strokes, and heart attacks by up to 80%.

SOLUTION

- **Connected & ready:** Patients register to hospitals with all of their health information and for instant and ready emergency access.
- **Real-time alerts:** Sends SMS to hospitals, caregivers, and ambulances instantly.
- **Fast and effective**: Detects emergencies—expected (pregnancy) and unexpected (falls)—and sends instant alerts.
- **Affordable & wearable:** Low-cost, user-friendly bracelet for low-resource areas.
- **Smart monitoring:** Tracks vital signs (heart rate, oxygen, temperature, motion) and prevents emergencies by alerting hospitals even before a fall occurs.

COMPONENTS

- Ardiono Nano
- MPU6050
- Micropush button
- GPS module(Neo-6M)
- GSM Module(SIM800A)
- MAX300: heartbeat and oximeter
- GY906: Temprature
- Casing and strap(Silicone/Fabric recycled plastic)

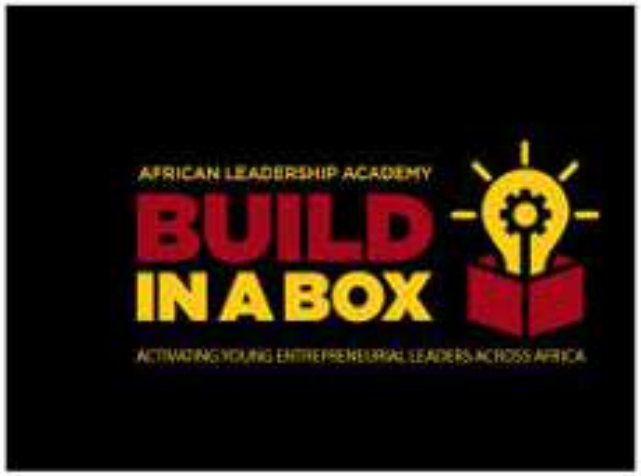
IMPACT

- **Reduce deaths by up to 80%:** Enables early detection and faster response to heart attacks, strokes, and pregnancy emergencies.
- **Cut response time by 60%:** Real-time GPS alerts connect patients to hospitals and ambulances instantly.
- **Protect 3× more high-risk patients:** Provides continuous monitoring for pregnant women, elderly, and chronically ill individuals.
- **Affordable for all**: Designed to cost under average cost of watches making life-saving tech accessible in low-resource regions.

FUTURE WORK

- Develop a community-driven wearable emergency bracelet for hospitals, clinics, and high-risk individuals.
- Enhance with smart sensors (BP, glucose, ECG) for accurate health monitoring.
- Build an integrated app linking hospitals, ambulances, and caregivers for rapid response.
- Expand nationwide partnerships with healthcare networks.
- Continuously improve features, reliability, and affordability through user feedback.

PAST ACHIEVEMENT





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System Hardware Overview

- The figure shows the two PCBs of the system: a top PCB (43 × 43 mm) and a bottom PCB (26 × 26 mm).
- the top PCB is positioned above the wrist (like a watch face) and handles processing and communication, while the bottom PCB is placed underneath on the strap/handle part, in contact with the skin, and contains the sensing components (temperature, motion, and SpO_2).
- Together, they form a compact, watch-style wearable device where sensing occurs at the bottom and control/communication happens on top.